Statistical Data Analysis Project

Practical Assignment 2

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Loading Packages

```
library(tidyverse)  library(grid)
library(lubridate)  library(psych)
library(ggplot2)    library(readr)
library(dplyr)      library(ggthemes)
library(flextable)  library(tidyr)
library(officer)    library(rlang)
library(gridExtra)  library(patchwork)
```

Loading The Data

```
setwd("C:/Users/flowe/OneDrive/Desktop/RMIT/Masters/Applied Analytics/Assignment 2/Possible Data")

setwd("C:/Users/llerr/OneDrive - RMIT University/RMIT_Semester_1_24/Applied Analytics/AA - Assignment 2/AA - Possible Data")

bike <- read.csv("SeoulBikeData.csv", check.names = F)
```

```
bike %>%
 summarize_all(class) %>%
 gather(variable, class)
```

```
##                       variable     class
## 1                         Date character
## 2            Rented Bike Count   integer
## 3                         Hour   integer
## 4            Temperature(\xb0C)   numeric
## 5                  Humidity(%)   integer
## 6             Wind speed (m/s)   numeric
## 7             Visibility (10m)   integer
## 8  Dew point temperature(\xb0C)   numeric
## 9      Solar Radiation (MJ/m2)   numeric
## 10                Rainfall(mm)   numeric
## 11               Snowfall (cm)   numeric
## 12                     Seasons character
## 13                     Holiday character
## 14             Functioning Day character
```

Data Wrangling, Factoring & Tidying

- Refactoring the data for ease of Wrangling.
- Date was formatted, data renaming was performed for clarity, and variables were converted to appropriate formats.
- Missing values were examined.
- Grouping, binning, and tidying steps were performed for further analysis.

```
bike <- setNames(bike,
c("Date","Rented.Bike.Count","Hour","Temperature","Humidity","Windspeed","Visibility","Dewpointtemperature","SolarRadiation","Rainfall","Snowfall","Seasons","Holiday","FunctioningDay"))

bike$Date <- as.Date(bike$Date, format = "%d/%m/%Y")
bike$Hour <- factor(bike$Hour, order = TRUE)
bike$Seasons <- factor(bike$Seasons)
bike$Holiday <- factor(bike$Holiday)
bike$FunctioningDay <- factor(bike$FunctioningDay)
bike$YearMonth <- format(as.Date(bike$Date), "%Y-%m")

bike_by_month <- bike %>%
 group_by(YearMonth) %>%
 summarise(Total.Bike.Count = sum(Rented.Bike.Count))

bike_by_month$FormattedMonth <- format(as.Date(paste0(bike_by_month$YearMonth, "-01")), "%b-%y")
```

Attribute Descriptions

- **Date:** Daily timestamp tracker.
- **Rented Bike Count:** Number of bikes rented per hour.
- **Temperature (°C):** Measures atmospheric heat.
- **Humidity (%):** Measures atmospheric moisture.
- **Wind Speed (m/s):** The rate of air movement measured in meters per second.
- **Visibility (10m):** Distance at which objects are visible, measured in units of 10 meters.
- **Dew Point Temperature (°C):** The temperature in Celsius at which air must be cooled for condensation (dew or frost).
- **Solar Radiation (MJ/m^2):** Sunlight intensity measured in megajoules per square meter.
- **Rainfall (mm):** Quantity of rainfall, measured in millimeters.
- **Snowfall (cm):** Quantity of snow, measured in centimeters.
- **Seasons:** Climate conditions for the given period (Winter, Spring, Summer, Autumn).
- **Holiday:** Indicates whether the observation falls on a public holiday.
- **Functioning Day:** Days when bike rentals were operational. Non-functioning days indicate 0 rentals throughout the day (e.g., when the rental service was closed).

Introduction

Data Collection & Information

The Seoul Bike Data Analysis contains continuous data on 24-hour rolling information of bikes rented between December 1, 2017, and November 30, 2018.

The data was sourced from the **UCI Machine Learning Repository**, an open-source platform. The rented bike count data was sourced from the company's internal database systems, while environmental factors, such as temperature and rainfall, were obtained from official weather records. Temporal data was recorded on a timestamp basis to ensure accuracy and consistency, with all data collected at the time of observation.

Analysis & Problem Statement

The aim of this analysis is to understand if the environmental, temporal, and operational factors that impact bike rentals.

Based on our observations, it is clear that weather data plays a significant role in the dataset.

This analysis aims to evaluate the usefulness and relevance of the weather data collected by addressing the following key question:

" What Factors Significantly Impact The Bike Rental Demand? "

The goal is to narrow the focus to a few key factors that are most impactful on demand, by thoroughly analyzing the dataset to gain a comprehensive understanding.

Steps In The Analysis:

Basic Observations and Arithmetic: Identify temporal and operational patterns and their relationship/impact on demand by using arithmetic operations.

Descriptive Statistics: Understand the distribution, ranges, and central tendency of continuous data to summarize key features by using summary statistics to describe the data patterns.

Data Visualization:

General Overview: Explore the data to identify significant factors impacting demand using grid of density plots to identify trends.

Deep Dive: Focus on the identified factors to analyze their effect in greater detail, setting the foundation for further analysis. Box plot, line trends and scatterplots were used here to understand the trends and correlations.

Chi-Square Analysis: Compare the categorical variable (Seasons) with binned values of rented bike counts to identify trends and associations by using chisq.test(), stacked bar plot, and heat map.

Simple Linear Regression: As atmospheric temperature influences and impacts other factors. A simple Linear Regression is performed with Rented Bike Count using lm() and diagnostic plots such as residual vs. fitted plots, Q-Q plots, and scale-location plots:

Multiple Linear Regression: Analyze Rented Bike Count vs. Temperature, Solar Radiation & Humidity Levels, to understand the combined effect of multiple variables on demand and summarize their relationships using lm() and diagnostic plots such as residual vs. fitted plots, Q-Q plots, and scale-location plots.

Data Source: https://archive.ics.uci.edu/dataset/560/seoul+bike+sharing+demand

Observations & Calculations

Summary of Bike Data Analysis

Metric	Value
Total Number Of Holidays	432
Percentage Of Holidays	4.93 %
Total Hours In A Year (365 Days * 24 Hours)	8760
Total Functioning Days	8465

Number of Days in Each Season

Seasons	# Days
Autumn	91
Spring	92
Summer	92
Winter	90

1. **Total Records:** There are 8,760 records in the dataset.
2. **Number of Holidays:** 432 holidays, which account for 5% of the total data. A significantly low level to impact analysis. Thus included.
3. **Functioning Days:** There are 8,465 functioning days and 295 non-functioning days. Non-functioning days were included in this analysis to take into account exogenous factors impacting demand.
4. **Season Days:** The number of days in each season is approximately equal.

```
# Summary of Bike Data Analysis

total_holidays <- sum(bike$Holiday == 'Holiday')
percentage_holidays <- (total_holidays / nrow(bike)) * 100
total_hours_year <- 365 * 24
total_functioning_days <- sum(bike$FunctioningDay == 'Yes’)

summary_stats <- data.frame(Metric = c("Total Number Of Holidays", "Percentage
Of Holidays", "Total Hours In A Year (365 Days * 24 Hours)", "Total Functioning
Days"), Value = c(total_holidays,paste0(round(percentage_holidays, 2), "
%"),total_hours_year,total_functioning_days))

knitr::kable(summary_stats, caption = "Summary of Bike Data Analysis")

# Number of Days in Each Season

season_days <- bike %>% group_by(Seasons) %>% summarize(# Days =
n_distinct(Date))

knitr::kable(season_days, caption = "Number of Days in Each Season")
```

Descriptive Statistics

This table provides key descriptive statistics for continuous variables in the bike rental dataset, summarizing the distribution, central tendency, spread, and completeness of the data.

	Min	Q1	Median	Q3	Max	Mean	SD	n	Missing
Rented_Bike_Count	0.0	191.0	504.50	1065.25	3556.00	704.60	645.00	8760	0
Temperature	-17.8	3.5	13.70	22.50	39.40	12.88	11.94	8760	0
Humidity	0.0	42.0	57.00	74.00	98.00	58.23	20.36	8760	0
Windspeed	0.0	0.9	1.50	2.30	7.40	1.72	1.04	8760	0
Visibility	27.0	940.0	1698.00	2000.00	2000.00	1436.83	608.30	8760	0
Dewpoint_Temperature	-30.6	-4.7	5.10	14.80	27.20	4.07	13.06	8760	0
Solar_Radiation	0.0	0.0	0.01	0.93	3.52	0.57	0.87	8760	0
Rainfall	0.0	0.0	0.00	0.00	35.00	0.15	1.13	8760	0
Snowfall	0.0	0.0	0.00	0.00	8.80	0.08	0.44	8760	0

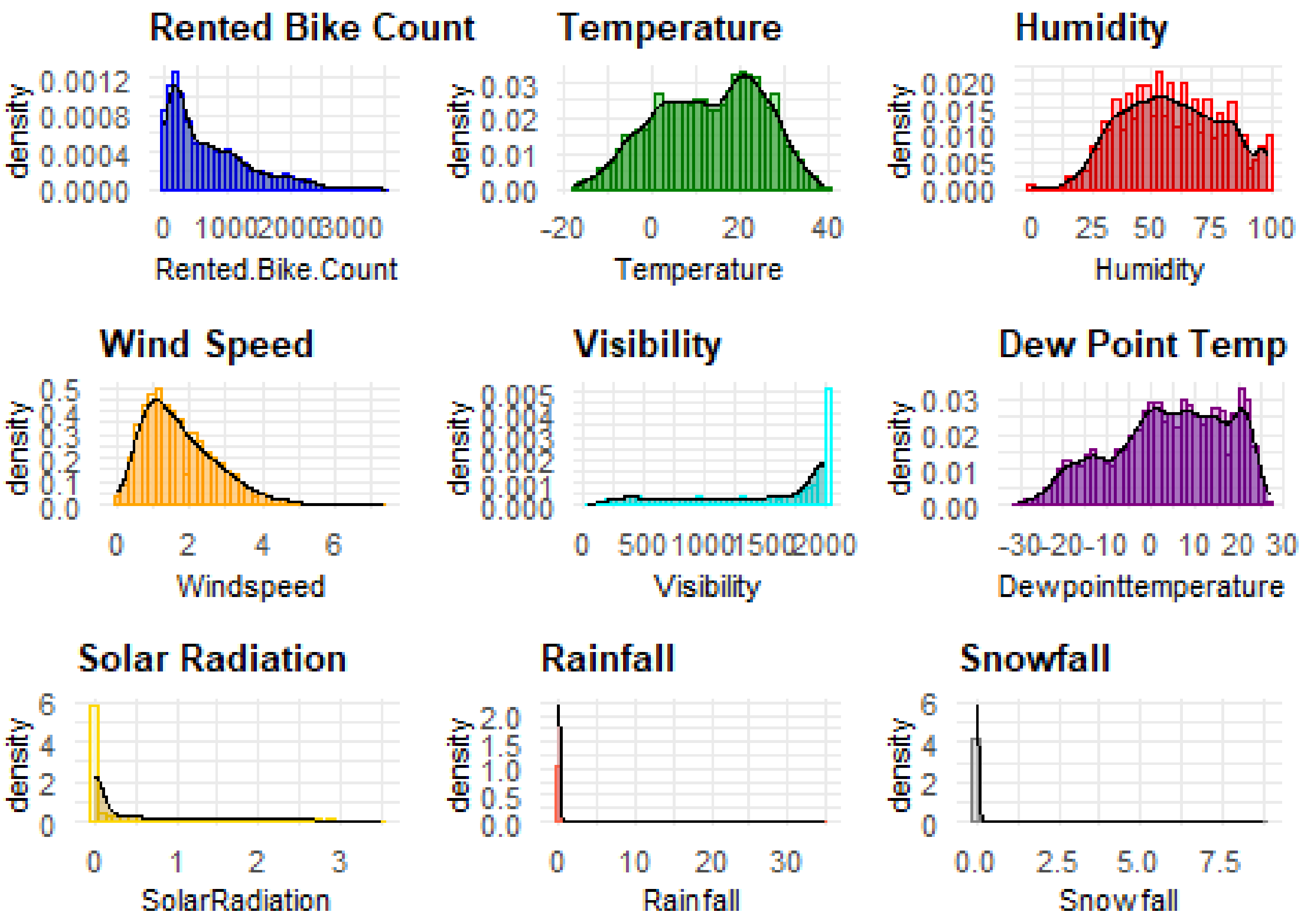
```r
summarize_stats <- function(x) {
 data.frame(
  Min = min(x, na.rm = TRUE),
  Q1 = quantile(x, probs = 0.25, na.rm = TRUE),
  Median = median(x, na.rm = TRUE),
  Q3 = quantile(x, probs = 0.75, na.rm = TRUE),
  Max = max(x, na.rm = TRUE),
  Mean = mean(x, na.rm = TRUE),
  SD = sd(x, na.rm = TRUE),
  n = sum(!is.na(x)),
  Missing = sum(is.na(x))
 )
}

summary_rented_bike <- summarize_stats(bike$Rented.Bike.Count)
summary_temperature <- summarize_stats(bike$Temperature)
summary_humidity <- summarize_stats(bike$Humidity)
summary_windspeed <- summarize_stats(bike$Windspeed)
summary_visibility <- summarize_stats(bike$Visibility)
summary_dewpointtemp <- summarize_stats(bike$Dewpointtemperature)
summary_solarradiation <- summarize_stats(bike$SolarRadiation)
summary_rainfall <- summarize_stats(bike$Rainfall)
summary_snowfall <- summarize_stats(bike$Snowfall)

summary_stats <- rbind(
 Rented_Bike_Count = summary_rented_bike,
 Temperature = summary_temperature,
 Humidity = summary_humidity,
 Windspeed = summary_windspeed,
 Visibility = summary_visibility,
 Dewpoint_Temperature = summary_dewpointtemp,
 Solar_Radiation = summary_solarradiation,
 Rainfall = summary_rainfall,
 Snowfall = summary_snowfall
)

knitr::kable(summary_stats, digits = 2, caption = "Summary Statistics for Bike
Data")
```

Summary Statistics for Bike Data

Rented Bike Count ranges from 0 to 3,556, with a mean of 704.6 and a standard deviation (SD) of 645, indicating high variability with many low-demand hours.
Temperature varies from -17.8°C to 39.4°C, averaging 12.88°C, reflecting seasonal effects.
Humidity ranges from 0% to 98%, with a mean of 58.23%, showing moderate moisture levels.
Windspeed has a low variability with a mean of 1.72 m/s and SD of 1.04.
Visibility spans from 270m to 2,000m, averaging 1,436.83m, suggesting generally clear conditions.

Dewpoint Temperature ranges between -30.6°C and 27.2°C, with a mean of 4.07°C, showing seasonal variation.
Solar Radiation has a right-skewed distribution, with most values near 0 and a mean of 0.57 MJ/m^2.
Rainfall and **Snowfall** are minimal, with medians of 0 and means of 0.15 mm and 0.08 cm respectively, indicating that precipitation is infrequent.
Missing Values are absent across all variables confirming complete hourly data for one year period.

Visual Analysis



Density Histograms of Individual Elements Reflecting The Trend of Continous Data

```
create_plot <- function(data, var, title, hist_fill, hist_color, density_fill) {
 ggplot(data, aes(x = !!sym(var))) +
  geom_histogram(aes(y = after_stat(density)), bins = 40, fill = hist_fill, color =
hist_color, alpha = 0.7) +   geom_density(alpha = 0.4, fill = density_fill) +
  labs(title = title) +  theme_minimal() +  theme(plot.title = element_text(size =
10, face = "bold"), axis.title = element_text(size = 8), axis.text = element_text(size
= 8))}

plots <- list(
 list(var = "Rented.Bike.Count", title = "Rented Bike Count",
    hist_fill = "#ADD8E6", hist_color = "#0000FF", density_fill = "#00008B"),
 list(var = "Temperature", title = "Temperature",
    hist_fill = "#90EE90", hist_color = "#008000", density_fill = "#006400"),
 list(var = "Humidity", title = "Humidity",
    hist_fill = "#FFB6C1", hist_color = "#FF0000", density_fill = "#8B0000"),
 list(var = "Windspeed", title = "Wind Speed",
    hist_fill = "#FFFFE0", hist_color = "#FFA500", density_fill = "#FF8C00"),
 list(var = "Visibility", title = "Visibility",
    hist_fill = "#E0FFFF", hist_color = "#00FFFF", density_fill = "#008B8B"),
 list(var = "Dewpointtemperature", title = "Dew Point Temp",
    hist_fill = "#DDA0DD", hist_color = "#800080", density_fill = "#4B0082"),
 list(var = "SolarRadiation", title = "Solar Radiation",
    hist_fill = "#FAFAD2", hist_color = "#FFD700", density_fill = "#B8860B"),
 list(var = "Rainfall", title = "Rainfall",
    hist_fill = "#F08080", hist_color = "#FF6347", density_fill = "#CD5C5C"),
 list(var = "Snowfall", title = "Snowfall",
    hist_fill = "#D3D3D3", hist_color = "#808080", density_fill = "#696969")
)

plot_list <- lapply(plots, function(p) {
 create_plot(bike, p$var, p$title, p$hist_fill, p$hist_color, p$density_fill)
})
grid.arrange(grobs = plot_list, ncol = 3,
      bottom = textGrob("Density Histograms of Individual Elements Reflecting
The Trend of Continous Data", gp = gpar(fontsize = 10, fontface = "bold"), hjust =
0.5))
```

Visual Analysis

Rented Bike Count: Right-skewed distribution with peak rentals between 0 – 1,000, fewer days with high rental counts.
Temperature: Indicates seasonality, with most values around 0°C and 15°C peaking at 20°C.
Humidity: Left-skewed, with heavy clustering above 25% humidity.
Windspeed: Left-skewed, suggesting low wind speeds.
Visibility: Heavily left-skewed, indicating high visibility is common; low visibility is rare.
Dewpoint Temperature: Right-skewed, suggesting warmer dew points and more humid conditions.
Solar Radiation: Left-skewed, indicating low solar radiation.
Rainfall: Sharp spike at the lower end, showing the absence of rainfall on most days.
Snowfall: Large spike at 0, indicating minimal measurable snowfall.

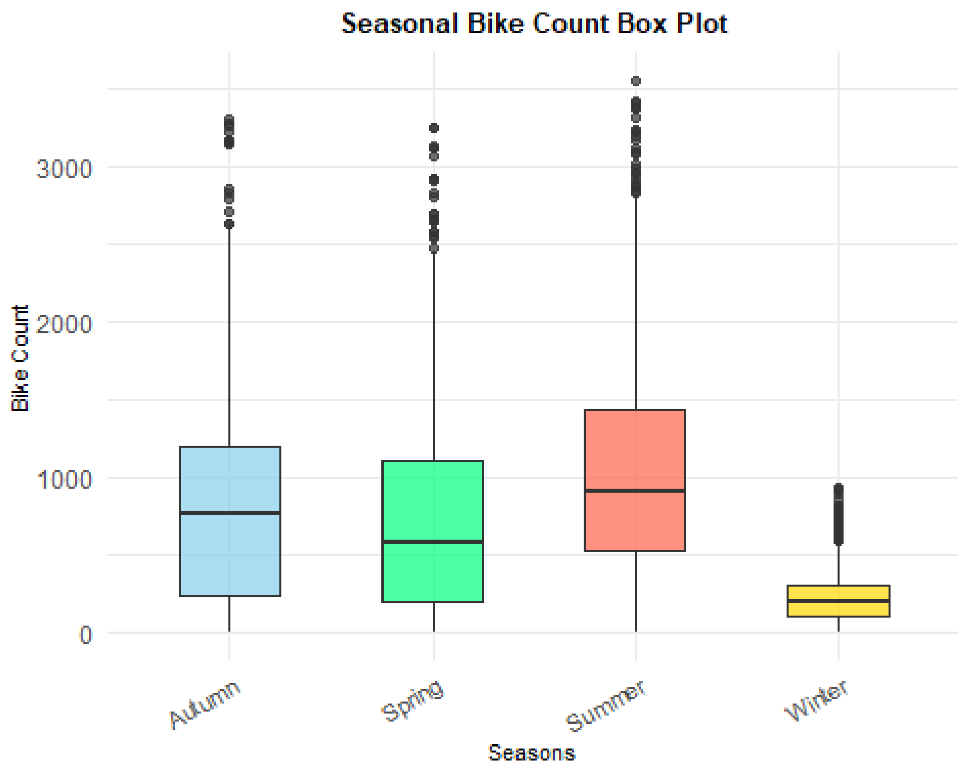
Approach

Based on the data above and the reasoning below, the factors temperature, humidity and solar radiation would be used for Regression Analysis. Additionally, due to the seasonality indicated by temperature, a Chi Square Analysis would be done with Seasons breaking down the Rented Bike Count data into Bin values.

Reasoning

High temperatures combined with high humidity and strong solar radiation make the weather feel hotter and more uncomfortable. Lower humidity and lower solar radiation can mitigate the intensity of hot weather, making it more comfortable. The interaction of temperature, humidity, and solar radiation during different seasons explains why the “feels-like” temperature can vary significantly from the actual air temperature. These factors together determine the overall perceived comfort of the weather, impacting bike rentals.

Seasonal Distribution of Bike Count



Summer shows the highest median bike rentals, with an interquartile range between 500 and 1,500 rental. Many outliers exceed 3,000 rentals, suggesting frequent peaks in demand due to favorable weather.

Spring & Autumn have relatively high rental counts, with overlapping interquartile ranges, indicating similar rental demand. Both seasons show fewer outliers compared to Summer.

Winter shows significantly lower demand, with a narrower interquartile range, most rentals falling below 500. Outliers rarely exceed 1,000 rentals, reflecting the impact of colder weather on bike rentals.

```
bike_count_bp <- ggplot(bike, aes(x = as.factor(Seasons), y = Rented.Bike.Count, fill =
Seasons)) +
 geom_boxplot(alpha = 0.7, width = 0.5) +
 scale_fill_manual(values = c("skyblue", "springgreen", "tomato", "gold")) +
 ggtitle("Seasonal Bike Count Box Plot") +
 xlab("Seasons") + ylab("Bike Count") +
 theme_minimal() +
 theme(
  plot.title = element_text(size = 10, face = "bold", hjust = 0.5),
  axis.title.x = element_text(size = 8),
  axis.title.y = element_text(size = 8),
  axis.text.x = element_text(size = 8, angle = 30, hjust = 1),
  legend.position = "none"
 )
bike_by_month$Season <- case_when(
 format(as.Date(paste0(bike_by_month$YearMonth, "-01")), "%m") %in% c("12", "01",
"02") ~ "Winter",
 format(as.Date(paste0(bike_by_month$YearMonth, "-01")), "%m") %in% c("03", "04",
"05") ~ "Spring",
 format(as.Date(paste0(bike_by_month$YearMonth, "-01")), "%m") %in% c("06", "07",
"08") ~ "Summer",
 TRUE ~ "Autumn"
)
season_colors <- c("Winter" = "skyblue", "Spring" = "springgreen", "Summer" =
"tomato", "Autumn" = "gold")

bike_by_month <- bike_by_month %>%
 mutate(Date = as.Date(paste0(YearMonth, "-01")))
print(bike_count_bp)
```

Seasonal Distribution of Bike Count

Total Bike Rentals by Month - Trend Analysis

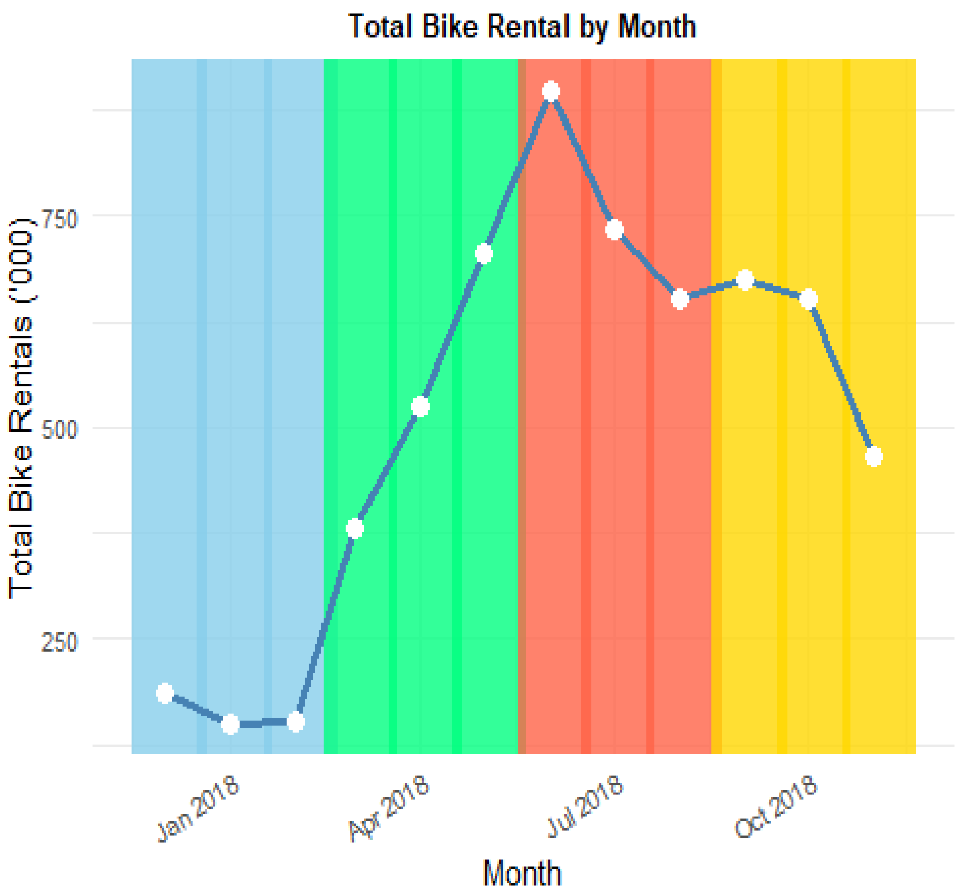
Winter (Dec-Feb): Lowest bike rentals, with a drop to under 200,000 rentals in January.

Spring (Mar-May): A significant increase in rentals, with demand rising sharply from February to May.

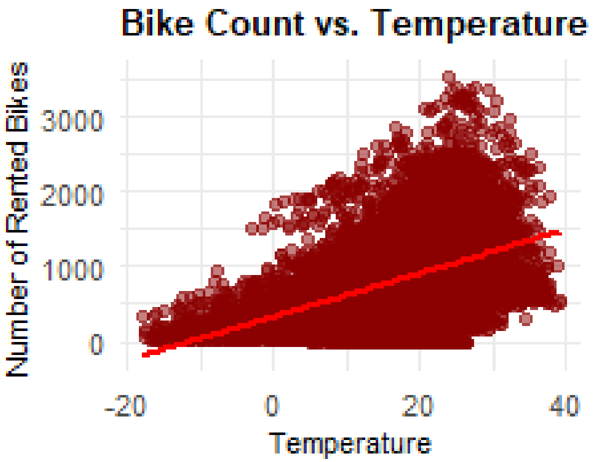
Summer (Jun-Aug): Peak rental activity, with July and August showing the highest numbers (over 800,000 rentals).

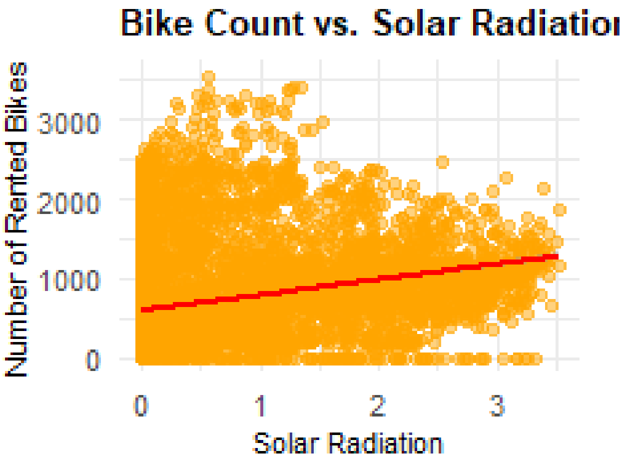
Autumn (Sep-Nov): A decline begins in September, with a more noticeable drop in October and November as the season transitions towards winter.
In Summer, with higher temperatures and solar radiation, we see a peak for bike rentals, while Winter's colder temperatures and higher humidity result in the lowest demand. Rentals rise in Spring with warming temperatures and decline in Autumn as conditions cool.

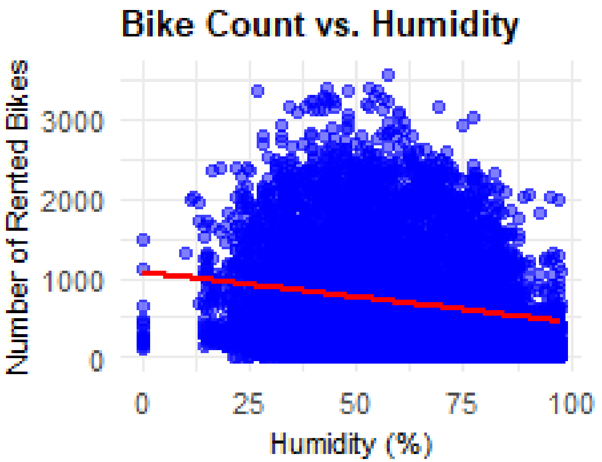
```
line_plot <- ggplot(bike_by_month, aes(x = Date, y = Total.Bike.Count / 1000)) +
 geom_rect(aes(xmin = Date - 15, xmax = Date + 20, ymin = -Inf, ymax = Inf, fill = Season), color
= NA, alpha = 0.8) + geom_line(color = "steelblue", linewidth = 1.2) + geom_point(color =
"white", size = 3) + scale_fill_manual(values = season_colors) +
 labs(title = "Total Bike Rental by Month",
  x = "Month",
  y = "Total Bike Rentals ('000)") + theme_minimal() + theme(
  plot.title = element_text(size = 10, face = "bold", hjust = 0.5),
  axis.text.x = element_text(size = 8, angle = 30, hjust = 1),
  axis.text.y = element_text(size = 8) ,
  legend.position = "none")

print(line_plot)
```



Correlation Analysis

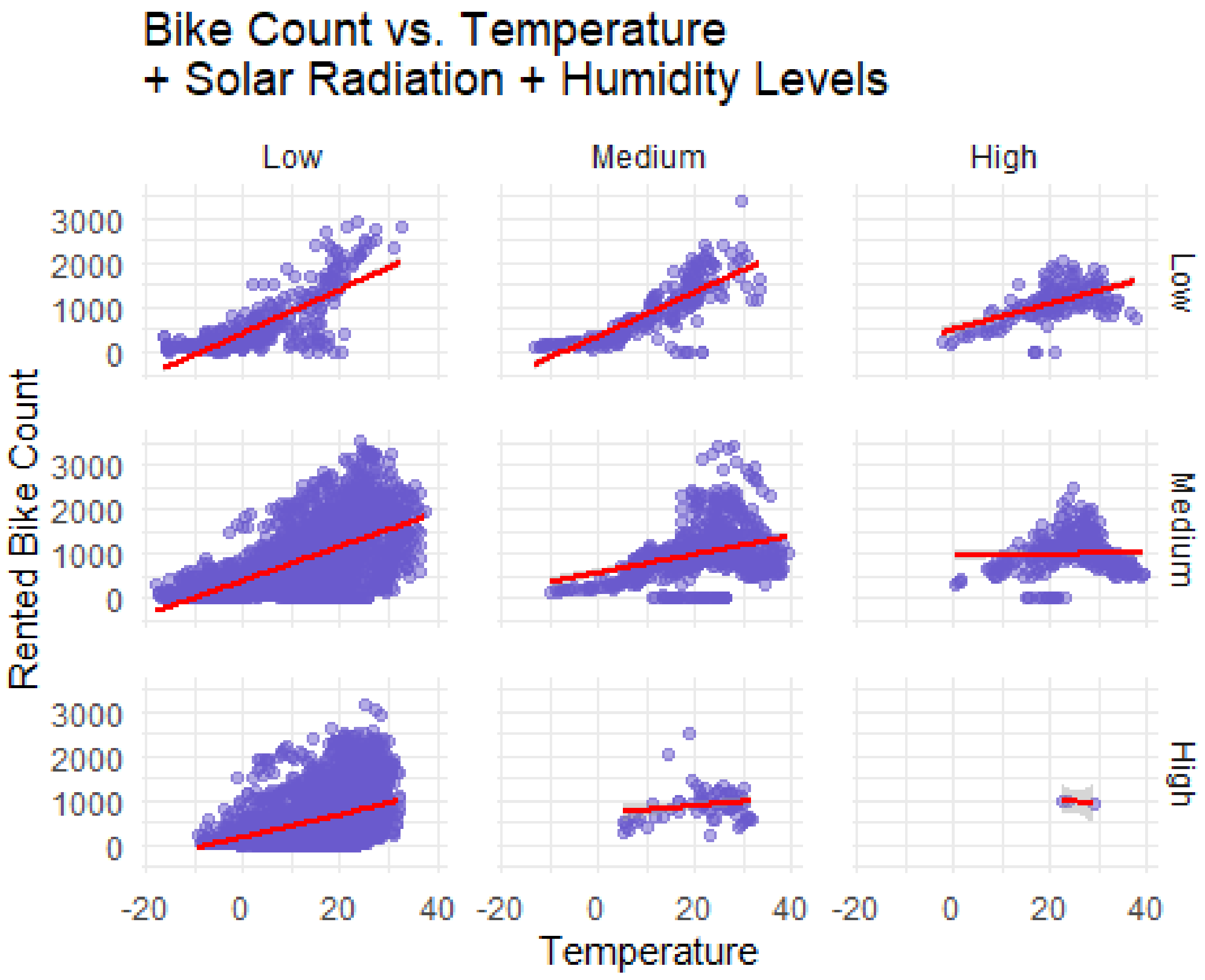






- **Bike Count vs. Temperature**: The scatter plot shows a positive linear relationship between temperature and bike rentals. As temperature rises, the number of bike rentals also increases, with a clear upward trend.
- **Bike Count vs. Solar Radiation**: A slight positive trend is observed, where bike rentals increase as solar radiation increases. The correlation is weaker than with temperature, indicating that other factors may play a stronger role.
- **Bike Count vs. Humidity**: The red regression line indicates a negative relationship between humidity and bike rentals. As humidity increases, bike rentals tend to decrease, suggesting that higher humidity may make biking less comfortable.

Correlation Analysis



Bike Count vs. Temperature + Solar Radiation + Humidity Levels

Temperature: Bike rentals generally increase with higher temperatures, as shown by the positive slope in most panels.

Humidity: Low humidity maintains a strong positive relationship between temperature and bike rentals. As humidity increases, the slope flattens, suggesting that high humidity reduces the positive impact of temperature on rentals.

Solar Radiation: Under low solar radiation, temperature still drives bike rentals significantly. With high solar radiation, the relationship is weaker, possibly indicating that too much heat or sunlight limits outdoor activity.

Bike demand increases with temperature, but high humidity and intense solar radiation dampen the positive effect of warmer temperatures on rentals.

Correlation Analysis

```
# Plot 1: Temperature vs. Bike Count
plot_temp <- ggplot(bike, aes(x = Temperature, y = Rented.Bike.Count)) +
 geom_point(alpha = 0.5, color = "darkred") +
 geom_smooth(method = "lm", se = TRUE, color = "red") +
 labs(title = "Bike Count vs. Temperature", x = "Temperature", y = "Number of
Rented Bikes") +
 theme_minimal() +
 theme(
  plot.title = element_text(size = 10, face = "bold"),
  axis.text = element_text(size = 8),
  axis.title = element_text(size = 8)
 )

# Plot 2: Solar Radiation vs. Bike Count
plot_radiation <- ggplot(bike, aes(x = SolarRadiation, y = Rented.Bike.Count)) +
 geom_point(alpha = 0.5, color = "orange") +
 geom_smooth(method = "lm", se = TRUE, color = "red") +
 labs(title = "Bike Count vs. Solar Radiation", x = "Solar Radiation", y = "Number of
Rented Bikes") +
 theme_minimal() +
 theme(
  plot.title = element_text(size = 10, face = "bold"),
  axis.text = element_text(size = 8),
  axis.title = element_text(size = 8)
 )
```

```
# Plot 3: Humidity vs. Bike Count
plot_humidity <- ggplot(bike, aes(x = Humidity, y = Rented.Bike.Count)) +
 geom_point(alpha = 0.5, color = "blue") +
 geom_smooth(method = "lm", se = TRUE, color = "red") +
 labs(title = "Bike Count vs. Humidity", x = "Humidity (%)", y = "Number of Rented
Bikes") +
 theme_minimal() +
 theme(
  plot.title = element_text(size = 10, face = "bold"),
  axis.text = element_text(size = 8),
  axis.title = element_text(size = 8)
 )

combined_plot <- plot_temp + plot_radiation + plot_humidity +
 plot_layout(ncol = 2, guides = "collect")
print(combined_plot)
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
```

```
# Bike Count vs. Temperature + Solar Radiation + Humidity Levels
bike$SolarRadiation_Binned <- cut(bike$SolarRadiation, breaks = 3, labels =
c("Low", "Medium", "High"))
bike$Humidity_Binned <- cut(bike$Humidity, breaks = 3, labels = c("Low",
"Medium", "High")
)
ggplot(bike, aes(x = Temperature, y = Rented.Bike.Count)) + geom_point(alpha
=0.5, color ="#6A5ACD") + geom_smooth(method ="lm", se =TRUE, color ="red") +
facet_grid(Humidity_Binned ~ SolarRadiation_Binned) + labs(title = "Bike Count
vs. Temperature + Solar Radiation + Humidity Levels",x = "Temperature",y =
"Rented Bike Count") + theme_minimal() + theme(panel.spacing = unit(1,"lines"))
```

Chi Square Analysis

Hypothesis Testing:

Null Hypothesis (H0): There is no association between the season and the bike count categories. In other words, bike rental counts are distributed independently of the seasons.
Alternative Hypothesis (H1): There is an association between the season and the bike count categories, meaning that bike rentals depend on the season.

Statistical Results:

With a **p-value < 2.2e-16**, we reject the null hypothesis.
There is significant evidence to suggest that bike rental counts are not independent of the seasons.
Seasonality plays a significant role in bike rentals, with Spring and Summer having a more balanced distribution across bike count categories.
Higher rentals are observed in the 1,000 - 1,999 and 2,000 - 2,999 ranges during Spring and Summer compared to Winter, where most rentals fall into the 0 - 999 category.
The X-squared = 1,276, suggesting a large deviation between the observed and expected counts.

Observed Bike Rental Counts by Season

Season	0-999	1000-1999	2000-2999	3000-3999
Autumn	1393	670	112	9
Spring	1559	546	97	6
Summer	1230	722	240	16
Winter	2160	0	0	0

Expected Bike Rental Counts by Season

Season	0-999	1000-1999	2000-2999	3000-3999
Autumn	1581	483	112	8
Spring	1599	488	113	8
Summer	1599	488	113	8
Winter	1564	478	111	8

Pearson's Chi-Square Test Results

X-squared	Degrees of Freedom	p-value
1276.16	9	< 2.22e-16

Seasons vs. Binned Bike Count

Chi Square Analysis

```
max_count <- ceiling(max(bike$Rented.Bike.Count) / 1000) * 1000

bike <- bike %>%
  mutate(Bike_Count_Category = cut(Rented.Bike.Count,
                    breaks = seq(0, max_count, by = 1000),
                    labels = paste0(head(seq(0, max_count, by = 1000), -1),
                            "-", tail(seq(0, max_count, by = 1000), -1) - 1),
                    include.lowest = TRUE))

table_seasons_bike <- as.data.frame(table(bike$Seasons,
bike$Bike_Count_Category))

chi_sq <- chisq.test(xtabs(Freq ~ Var1 + Var2, data = table_seasons_bike))

if (any(chi_sq$expected < 5)) {
  cat("Warning: Some expected counts are < 5. Chi-squared approximation
may be inaccurate.\n")
}

obs_df <- table_seasons_bike %>%
  pivot_wider(names_from = Var2, values_from = Freq, values_fill = 0) %>%
  rename(Season = Var1)

exp_df <- as.data.frame(as.table(chi_sq$expected)) %>%
  mutate(Freq = round(Freq)) %>%
  pivot_wider(names_from = Var2, values_from = Freq, values_fill = 0) %>%
  rename(Season = Var1)
```

```
chi_squared_results <- data.frame(
  Metric = c("X-squared", "Degrees of Freedom", "p-value"),
  Value = c(round(chi_sq$statistic, 2), chi_sq$parameter,
format.pval(chi_sq$p.value, digits = 5))
) %>%
  pivot_wider(names_from = Metric, values_from = Value)

knitr::kable(obs_df, caption = "Observed Bike Rental Counts by Season",
row.names = FALSE)
knitr::kable(exp_df, caption = "Expected Bike Rental Counts by Season",
row.names = FALSE)
knitr::kable(chi_squared_results, caption = "Pearson's Chi-Square Test
Results", row.names = FALSE)


heatmap_data_seasons <- bike %>% group_by(Seasons, Bike_Count_Category) %>%
summarise(Count = n(), .groups = "drop")
# Stacked Plot
plot_seasons_stacked <- ggplot(bike, aes(x = Seasons, fill = Bike_Count_Category)) +
geom_bar(position = "stack", width = 0.6) + scale_fill_brewer(palette = "Set3") +
labs(title = "Stacked Bar", x = "Season", y = "Bike Rentals", fill = "Count Category") +
theme_minimal() + theme(plot.title = element_text(size = 10, hjust = 0.5), axis.text.x =
element_text(angle = 45, hjust = 1), plot.margin = margin(5, 5, 5, 5))
# Heatmap
plot_seasons_heatmap <- ggplot(heatmap_data_seasons, aes(x =
Bike_Count_Category, y = Seasons, fill = Count)) + geom_tile(color = "white") +
scale_fill_gradient(low = "lightyellow", high = "red") + labs(title = "Heatmap", x = "Bike
Count Category", y = "Season", fill = "Count") + theme_minimal() + theme(plot.title =
element_text(size = 10, hjust = 0.5), axis.text.x = element_text(angle = 45, hjust = 1),
plot.margin = margin(5, 5, 5, 5))

print(plot_seasons_stacked)
print(plot_seasons_heatmap)
```

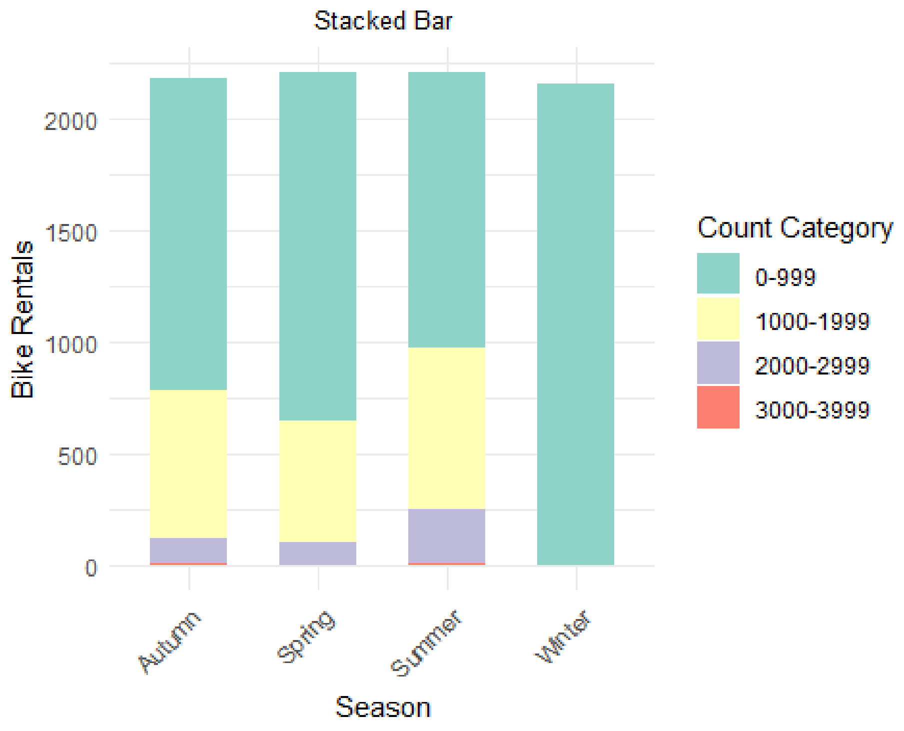
Chi Square Analysis Seasons vs. Bike Count

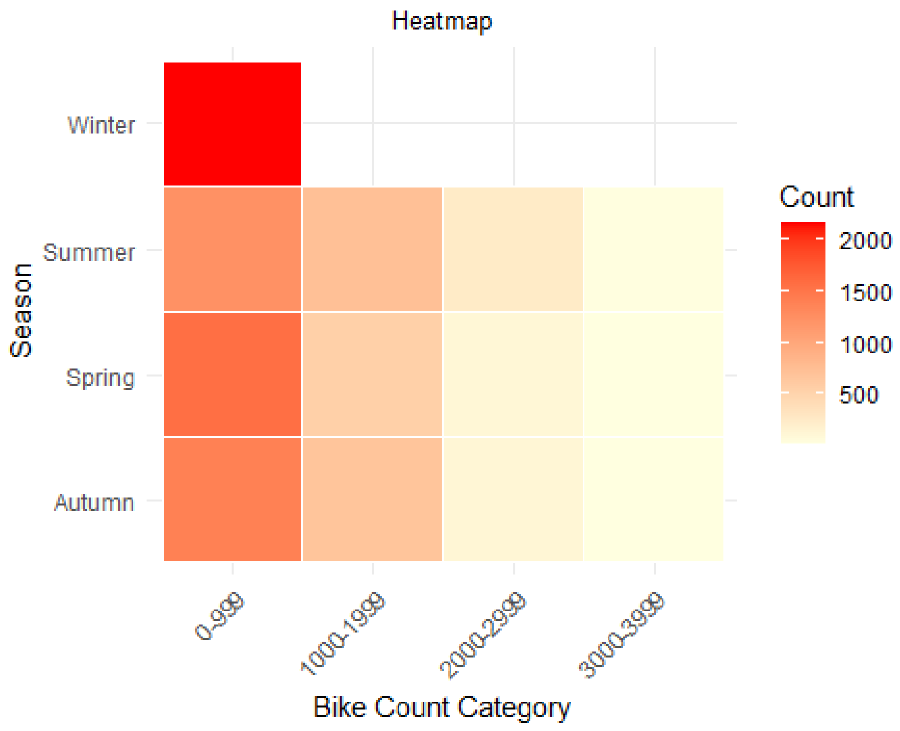
Bar Plot

The bar plot shows the distribution of bike rentals across different seasons and rental count categories. Winter shows the majority of rentals in the 0 - 999 category, indicating lower demand. Summer and Spring show higher rentals in the 1,000 - 1,999 and 2,000 - 2,999 categories, reflecting increased demand in warmer months.

Heatmap

The heatmap provides a detailed view of rental distribution, with darker regions representing higher concentrations of rentals. Winter has the lowest bike counts, while Spring and Summer exhibit more varied distribution of rentals, further supporting the hypothesis that bike demand is higher in warmer months.

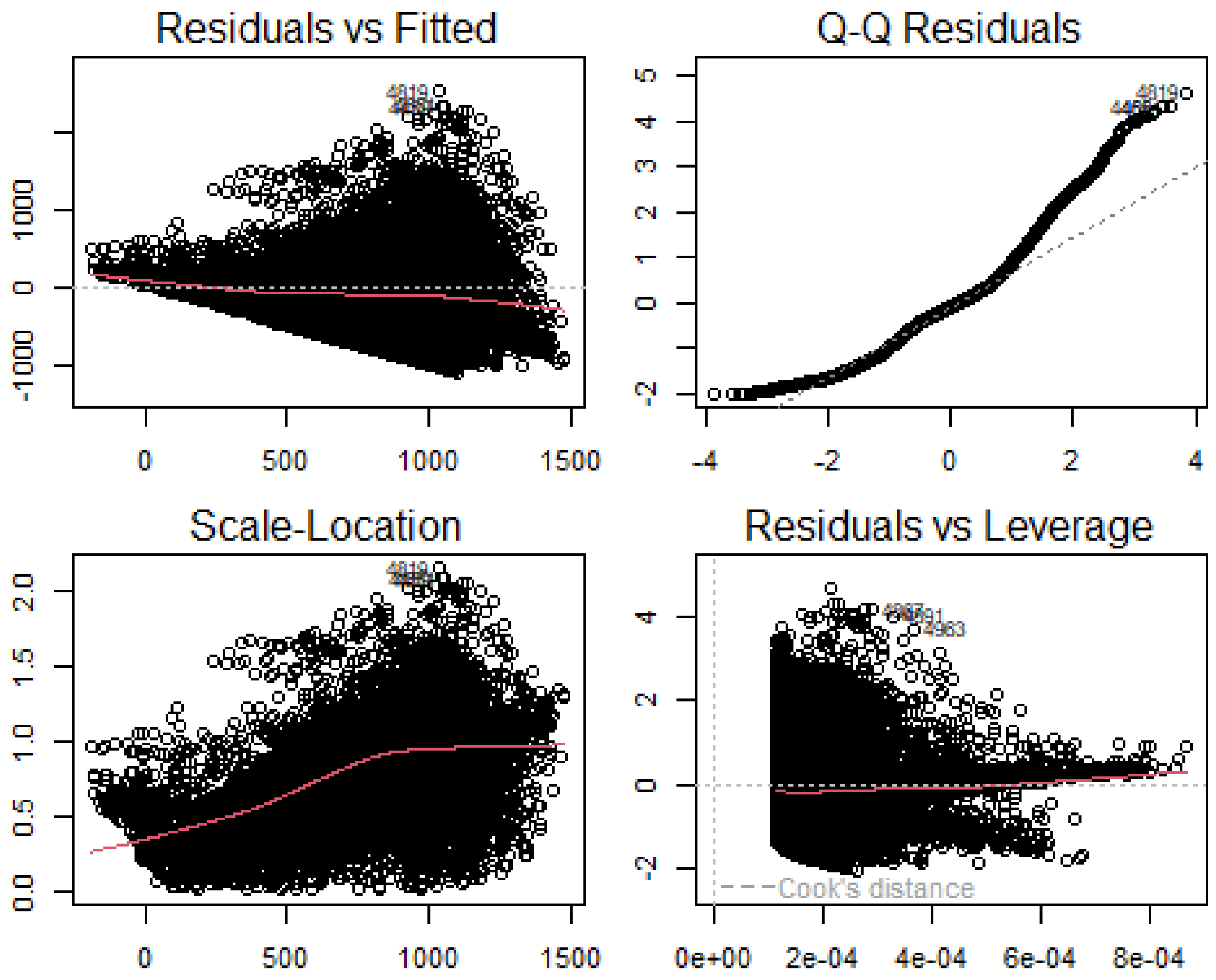




The analysis shows a significant association between seasons and bike rentals, with higher demand in warmer seasons like Spring and Summer. Winter has the lowest rental counts, while the data supports that seasonality strongly influences bike rental patterns.

Simple Linear Regression and Correlation

Simple Linear Regression aims to understand the relationship between variables to make accurate predictions. For this analysis, we're breaking down the major factor that impact weather, temperature to understand its relationship with the rented bike count (dependent variable).



Call:
lm(formula = Rented.Bike.Count ~ Temperature, data = bike)

Residuals:

Min	1Q	Median	3Q	Max
-1100.60	-336.57	-49.69	233.81	2525.19

Coefficients:

	Estimate	Std. Error	t value	Pr(>|t|)	
(Intercept)	329.9525	8.5411	38.63	<2e-16	***
Temperature	29.0811	0.4862	59.82	<2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Residual standard error: 543.5 on 8758 degrees of freedom
Multiple R-squared: 0.29, Adjusted R-squared: 0.29
F-statistic: 3578 on 1 and 8758 DF, p-value: < 2.2e-16

```
model_temp <- lm(Rented.Bike.Count ~ Temperature, data = bike)
summary(model_temp)
par(mfrow = c(2, 2), mar = c(2, 2, 2, 1) + 0.1, cex = 0.7)
plot(model_temp)
```

Rented Bike Count vs. Temperature

Simple Linear Regression and Correlation

Null Hypothesis (H0): The slope of the regression line is zero, meaning that temperature has no effect on bike rentals.
Alternative Hypothesis (H1): The slope of the regression line is not zero, meaning that temperature has a significant effect on bike rentals.

Statistical Explanation:

Intercept: When temperature is 0°C, approximately 330 bikes are expected to be rented.Represents the baseline number of rentals when the temperature is zero.
Temperature Coefficient: For every 1°C increase in temperature, the number of bikes rented increases by about 29 bikes.Shows a positive relationship between temperature and bike rentals.
Standard Error:
Intercept: 8.5, indicating a low prediction error.
T-Values:
Intercept: t = 38.63
Temperature: t = 59.82
Both values are very large, indicating a strong and statistically significant relationship with bike rentals.
P-Values: P-values for both the intercept and temperature are extremely small (p = 2e-16).We reject the null hypothesis, concluding that temperature significantly affects bike rentals.
R-Squared: R-squared = 0.29, meaning that about 29% of the variation in bike rentals is explained by temperature.Other factors account for the remaining 71% of the variation.
Adjusted R-Squared: Adjusted R-squared = 0.29, confirming that temperature is the only predictor in the model.
F-Statistic: F-statistic = 3578 with a p-value < 2.2e-16, suggesting that the model as a whole is statistically significant. Temperature has a significant effect on bike rentals.

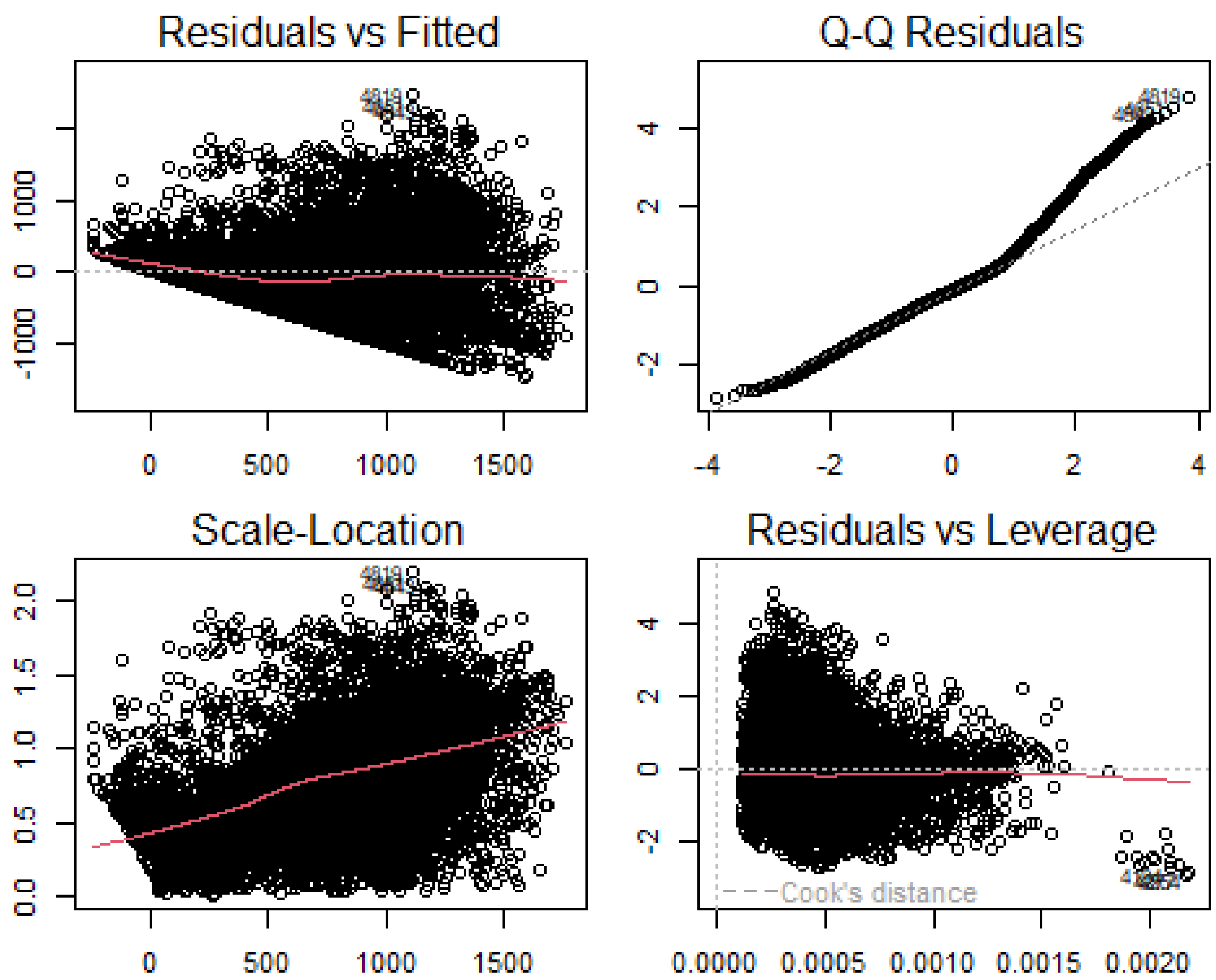
1. **Residuals vs. Fitted Plot:** The red line is not perfectly horizontal, and the spread of residuals increases as fitted values increase, suggesting heteroscedasticity (non-constant variance of residuals).
2. **Q-Q Plot:** The points curve away from the 45-degree line, suggesting non-normality in the residuals, especially in the tails.
3. **Scale-Location Plot:**The red line trends upward, indicating heteroscedasticity. Residuals tend to have larger variance at higher fitted values.
4. **Residuals vs. Leverage Plot:** Shows no extreme outliers, but some points with higher leverage may have a small influence on the model.

Summary:

The regression model shows a significant positive relationship between temperature and bike rentals, with each 1°C increase leading to about 29 more rentals. The R-squared value of 0.29 indicates that temperature explains a moderate portion of the variation in rentals, but other factors account for the remaining variation. There are signs of heteroscedasticity and non-normal residuals, though no extreme outliers are present.

Multiple Linear Regression and Correlation

Rented Bike Count vs. Temperature + Solar Radiation + Humidity



Call:
lm(formula = Rented.Bike.Count ~ Temperature + SolarRadiation + Humidity, data = bike)

Residuals:

Min	1Q	Median	3Q	Max
-1449.51	-315.11	-49.32	220.68	2446.23

Coefficients:

	Estimate	Std. Error	t value	Pr(>|t|)	
(Intercept)	985.3313	20.5296	48.00	<2e-16	***
Temperature	34.7780	0.5253	66.20	<2e-16	***
SolarRadiation	-99.6094	8.0397	-12.39	<2e-16	***
Humidity	-11.5426	0.3250	-35.51	<2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 506.1 on 8756 degrees of freedom

Multiple R-squared: 0.3845, Adjusted R-squared: 0.3843

F-statistic: 1824 on 3 and 8756 DF, p-value: < 2.2e-16

```
model_multiple_humidity <- lm(Rented.Bike.Count ~ Temperature
+ SolarRadiation + Humidity, data = bike)
summary(model_multiple_humidity)
par(mfrow = c(2, 2), mar = c(2, 2, 2, 1) + 0.1, cex = 0.7)
plot(model_multiple_humidity)
```

Multiple Linear Regression and Correlation

Null Hypothesis (H0): The slopes of the regression line for temperature, solar radiation, and humidity are zero, meaning that none of these variables have an effect on bike rentals.
Alternative Hypothesis (H1): At least one of the slopes is not zero, meaning that at least one of these factors (temperature, solar radiation, or humidity) has a significant effect on bike rentals.

Statistical Explanation:
Intercept: The intercept is 985.33, meaning that when temperature, solar radiation, and humidity are all zero, approximately 985 bikes are expected to be rented.
Temperature Coefficient: The coefficient for temperature is 34.78. Every 1°C increase results in about 34.78 more bikes rented, holding other factors constant. Indicating a positive relationship between temperature and bike rentals.
Solar Radiation Coefficient: The coefficient for solar radiation is -99.61. Every 1 unit increase in solar radiation (measured in MJ/m^2) decreases bike rentals by about 99.61. Suggests over-exposure to sunlight may reduce outdoor activities like biking.
Humidity Coefficient: The coefficient for humidity is -11.54. For every 1% increase in humidity, the number of bikes rented decreases by about 11.54.Confirms a negative relationship between humidity and bike rentals.
Standard Errors:
Temperature: 0.53
Solar Radiation: 8.04
Humidity: 0.33
Small values suggest precise estimates.
T-Values:
Temperature: t = 66.20
Solar Radiation: t = -12.39
Humidity: t = -35.51
Large t-values indicate all predictors are statistically significant.

P-Values: All predictors have p-values < 2e-16. We reject the null hypothesis, meaning temperature, solar radiation, and humidity significantly affect bike rentals.
R-Squared: The R-squared value is 0.3845, meaning 38.45% of the variation in bike rentals is explained by temperature, solar radiation, and humidity. Other factors account for the remaining 61.55% of the variation.
P-Values: All predictors have p-values < 2e-16. We reject the null hypothesis, meaning temperature, solar radiation, and humidity significantly affect bike rentals.
R-Squared: The R-squared value is 0.3845, meaning 38.45% of the variation in bike rentals is explained by temperature, solar radiation, and humidity. Other factors account for the remaining 61.55% of the variation.
Adjusted R-Squared: The adjusted R-squared value is 0.3843, close to the regular R-squared.
F-statistic: F-statistic = 1,824 with p-value < 2.2e-16. Indicates the overall model is statistically significant.

1. **Temperature:** Across most panels, the red trendline indicates a positive relationship between temperature and bike rentals. Higher temperatures are associated with more rentals, especially in low and medium humidity.
2. **Humidity:** The slope becomes less steep with increasing humidity. High humidity weakens the effect of temperature on rentals.
3. **Solar Radiation:** The effect is less pronounced but low solar radiation shows stronger correlations between temperature and rentals.
4. **Extreme Cases:** In high humidity and high solar radiation conditions, bike rentals are lower, and the temperature effect is nearly flat.

Summary:
Temperature has a strong positive relationship with bike rentals, while solar radiation and humidity have negative relationships. For every 1°C increase, about 35 more bikes are rented. For every 1 unit increase in solar radiation, rentals decrease by about 100 bikes. For every 1% increase in humidity, rentals decrease by about 11 bikes. The R-squared value (0.3845) indicates these factors explain 38.45% of the variation in bike rentals.

Conclusion

The analysis indicates that temperature positively influences bike rental demand, with higher temperatures correlating with increased rentals. In contrast, solar radiation and humidity have negative effects, where higher levels are associated with a decrease in bike rentals. Mild sunny days with moderate temperatures but not excessively high solar radiation, and low humidity create the most favorable conditions for bike rentals, particularly during spring and summer seasons. However, substantial variability remains unexplained, suggesting the potential influence of other factors such as social behavior, work schedules, or special events. Non-functioning days were also included in the analysis to account for the possibility of these influences.

Reference

1. **Kaggle Reference:**

- Alhammad, Q., 2021. Predict bike sharing demand using regression models. Kaggle. Available at: https://www.kaggle.com/code/qasimalhammad/ibm-data-science-with-r-capstone-project#predict-bike-sharing-demand-using-regression-models [Accessed 13 Oct. 2024].

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3. **RPubs Article:**

- April, Y., 2021. Seoul Bike Sharing Demand Analysis. RPubs. Available at: https://rpubs.com/yonnapril/860251 [Accessed 13 Oct. 2024].

4. **AI-Assisted Code and Language Editing:**

- OpenAI, 2024. AI-assisted code and language enhancement [Online]. Available at: https://openai.com [Accessed 13 Oct. 2024]. AI was utilized to assist in editing code and refining language throughout this project, ensuring precision in technical content and clarity in communication.